

Muhammad Hamza

AI & WEB DEVELOPER

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PORTFOLIO portfolio-lime-rho-96.vercel.app

SUMMARY

Computer Science student and developer who ships AI-powered, full-stack products end to end — from React/TypeScript interfaces to Python and Node back ends, with applied work in retrieval-augmented generation (RAG) and LLM-integrated features. I learn by building and deploying real apps.

SELECTED PROJECTS

Synapse — AI Study OS

React · FastAPI · RAG · Three.js

A retrieval-augmented “Study OS” that ingests a user’s own documents into a vector store and answers questions grounded in their material, with recalled concepts rendered as an explorable 3D Mind Palace.

synapse-eight-ruby.vercel.app · github.com/MuhammadHamza2210/Synapse

Orion — Real-Time Stock Dashboard

React · FastAPI · WebSockets · AI

Streams live prices for 50 tickers over WebSockets with interactive charts, plus on-demand LLM analysis that explains any stock’s recent movement in plain language.

orion-nine-wine.vercel.app · github.com/MuhammadHamza2210/Orion

CodeScan AI — Code Plagiarism Detector

Python · ML · CodeBERT

Combines four engines — AST diffing, a trained ML model, winnowing fingerprints and CodeBERT embeddings — for structure-aware similarity scoring that resists renaming and reformatting.

github.com/MuhammadHamza2210/CodeScan-AI

Also built: LootLo (full-stack e-commerce, Flask/SQLite) · WildMind AI (Gemini-powered encyclopedia) · WeatherScope (Python data viz).

SKILLS

Languages Python · TypeScript · JavaScript · SQL

AI / ML RAG & LLM apps · Prompt engineering · LLM APIs (Gemini, Claude, Ollama) · embeddings & vector search

Frontend React · Vite · Tailwind CSS · Three.js / R3F · Framer Motion

Backend Node.js · Express · FastAPI · Flask · REST APIs · WebSockets

Tools Git & GitHub · SQLite · Streamlit · Vercel

EDUCATION

BS Computer Science — Bahria University, Karachi

2024 – 2027 (expected)

CGPA **3.81 / 4.0** — through 5th semester

Relevant coursework: Data Structures & Algorithms · Databases · OOP · Software Engineering · Operating Systems